PS Labs: IdentityIQ LogiPlex Connector User Guide

LogiPlex Connector: IdentityIQ 7.2+

This document explains the inner workings, installation and use of the LogiPlex Connector. The LogiPlex Connector is a special connector that provides functionality like the standard Logical connector and multiplex applications.

Document Revision History

Revision Date	Written/Edited By	Comments
April 2018	Menno Pieters	Initial Release
March 2019	Menno Pieters	Adapter mode added
October 2019	Menno Pieters	Several improvements based on customer deployments
July 2020	Menno Pieters	Cloud Gateway Support, PS Labs update, Deployment update
March 2021	Menno Pieters	Added “do’s and don’ts”.

Table of Contents

LogiPlex Connector Overview

This document explains what the LogiPlex connector is, its inner workings, installation and usage. The LogiPlex Connector is a special connector that provides functionality which is a mixture of the standard Logical connector and multiplex applications.

1. Introduction

The LogiPlex Connector is a possible alternative for logical applications, using features of the multiplex application type. The name of this connector is composed of the connector types that inspired the creation: the logical and the multiplex connector types.

From the *Direct Connectors Administration and Configuration Guide*:

> *The SailPoint Logical Connector is a read only connector developed to create objects that function like applications, but that are actually formed based on the detection of accounts from other, or tier, applications in existing identity cubes.*
>
> *For example, you might have one logical application that represents three other accounts on tier applications, an Oracle database, an LDAP authorization application, and a custom application for internal authentication. The logical application scans identities and creates an account on the logical application each time it detects the three required accounts on a single identity.*
>
> *You can then use the single, representative account instead of the three separate accounts from which it is comprised for certification, reporting, and monitoring.*

The Logical Connector has a number of performance drawbacks: it requires information to be aggregated and then iterates over identities to find suitable matches. This is often a very time-consuming task. The LogiPlex Connector tries to overcome these issues by processing logical applications on the fly, during aggregation. This approach allows the use of performance-enhancing connector features like partitioning, optimized aggregation and delta aggregation.

The Logical Connector acts as a wrapper around any normal connector. It can operate in two modes:

1. **Classic Mode:** It must be set up to point to a pre-configured application and will use all features of the target application for aggregation and provisioning. It uses two additional rules to process information when aggregating and provisioning account or group information.
2. **Adapter Mode:** an existing, pre-configured application is modified to use the LogiPlex adapter to wrap the normally used adapter. The same additional rules that are used in classic mode are still used to control the LogiPlex-behavior.

When aggregating, account and group information is enriched with additional attributes to indicate the name of the logical sub-application. This mechanism is the same as used for so called multiplex applications.

A possible downside of the LogiPlex Connector is that it may in some cases be a little harder to set up than a standard Logical Connector. The LogiPlex Connector requires more BeanShell coding to work than most Logical Connector configurations. For the Logical Connector the tiers can be configured by just selecting the relevant entitlements. Furthermore, the LogiPlex Connector is less suitable for combining information from multiple tiers, although the aggregation split rule can be used to perform side lookups.

2. Application Details

2.1. Application Type

The application type is “LogiPlex Connector” in classic mode or any other standard or custom application type in adapter mode.

3. Terminology

In this document, we will use three different terms for the application definitions involved in the aggregation and provisioning of LogiPlex data.

3.1. Master Application

The master application is a standard application, like a directory server (LDAP, Active Directory), a database or delimited file application. This application needs to be set up first and proven to be working correctly.

3.2. Main Application

The main application is the base LogiPlex application. The *Main* Application will be configured to point to the *Master* Application and use the connector defined in the *Master* Application to perform the actual work.

3.3. Sub Application

A Sub Application is any application derived from the *Main* Application, based on information read from accounts. Unless the aggregation task is configured to prevent this, Sub Applications will be generated on the fly, while aggregating data.

A Sub Application will have the Main Application defined as a Proxy.

3.4. Inner Working

The idea behind this connector is that we would like to use multiplex-like behavior to achieve the logical account grouping that a Logical Connector provides: sub-applications of a master application. As an

example, consider two sets of groups in a directory server giving access to a web application and a server application.

A multiplex application only allows an account to be redirected to a single sub-application, while we would like the same account to be related to the web application as well as the server application, listing only the relevant group memberships.

The LogiPlex Connector features a rule option that allows inspection of a freshly read account or group object (ResourceObject) and returns one or more slightly modified instances of this object in a map.

The LogiPlex Connector uses a special iterator class that wraps the iterator of the original connector but uses the split rule to retrieve the sub-application accounts.

3.4.1. Classic Mode

In the classic mode, the master application and main application are two different application definitions. The master application is set up and must be confirmed working. Then a second definition is created of the type “LogiPlex Application” which is pointed to the pre-existing master application.

Parts of the master application’s XML definition, like the provisioning policies, need to be copied over to the main application. The basics and rules of the LogiPlex main application can be configured in the user interface.

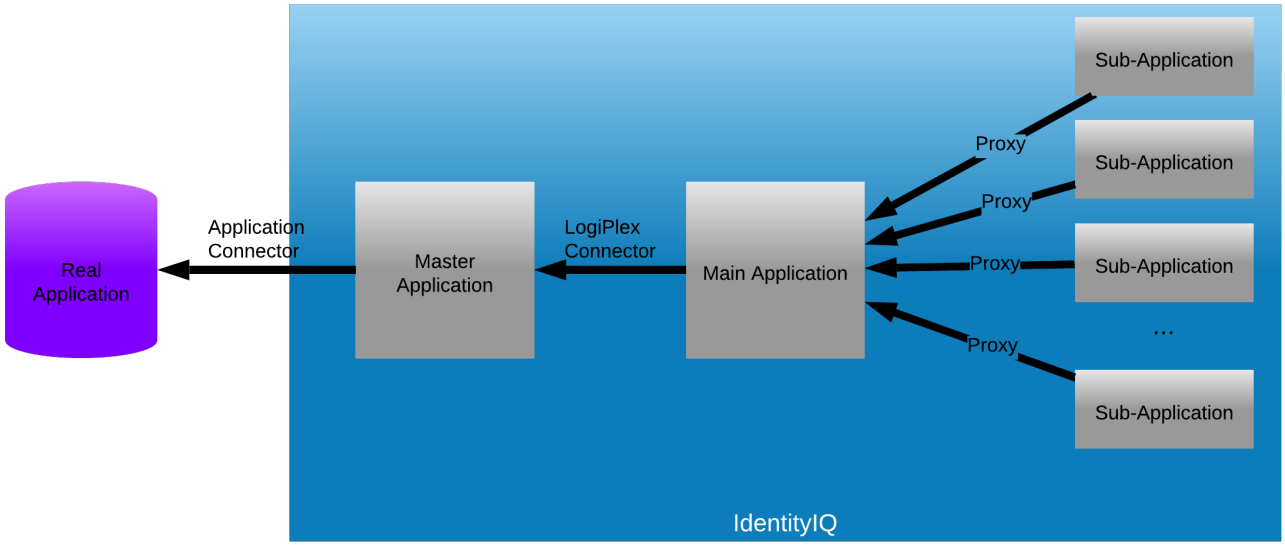


Figure 1: Relations between application definitions and connectors in “classic mode”

3.4.2. Adapter Mode

In the adapter mode, the master and main application become effectively one application definition. The master application is set up and must be confirmed working. Once that is done, a few changes need to be made to the application definition's XML:

- The real connector class is replaced by the class of the LogiPlex connector,
- The real connector class name is placed in a special attribute used by the LogiPlex connector,
- The aggregation (split) rule must be specified in the attributes map,
- Optionally a provisioning (merge) rule can be specified.

Adapter mode requires more manual changes to the XML definition but allows the master application to be updated with aggregation statistics and other information necessary for subsequent aggregations, especially when delta aggregation is used.

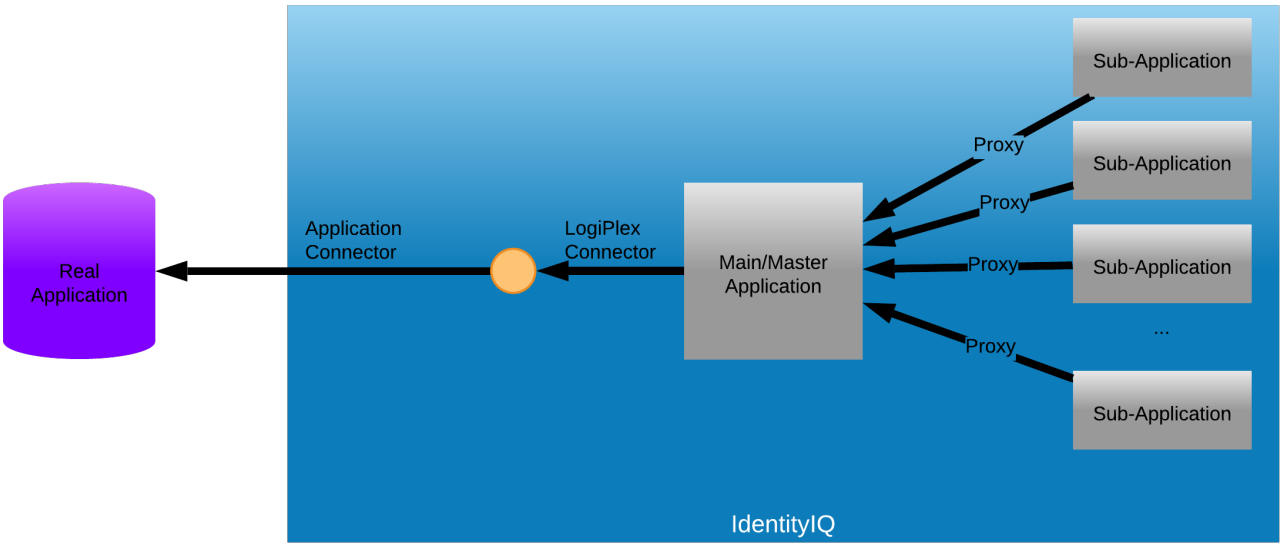


Figure 2: Relations between application definitions and connectors in "adapter mode"

Installation and Configuration – Classic Mode

4. Implementation Overview

The implementation of the LogiPlex connector in classic mode require the following steps:

- Deployment of the connector and supporting files and configurations,
- Set up the master application and confirm connectivity,
    - Test the configuration and schema
- Define and configure a LogiPlex application:
    - Set the master application,
    - Optionally define a prefix for auto-generated sub-applications,
    - Define a split rule to determine which entitlements belong to which sub-application
    - Optionally define a merge rule
    - For provisioning: configure a provisioning on the main application
    - For generated sub-applications:
        - Manually copy configuration options,
        - Manually copy and update the provisioning policy.

5. Installation Files

5.1. Services Standard Deployment (6.x and below)

If you are deploying IdentityIQ 7.2 or greater with the Services Standard Deployment (SSD), the LogiPlex Connector will be deployed by default, and there are no extra steps to take to install it. Deployment is controlled by this property in the `build.properties` file:

```
deployLogiPlexConnector=true
```

Setting it to false will prevent the connector being deployed.

The IdentityIQ LogiPlex Connector consists of the following Java class files:

Filename	Description
`LogiPlexConnector.java`	Main LogiPlex Connector Java class

The configuration files are:

Filename	Description
`ConnectorRegistry-LogiPlexConnector.xml`	Connector registry merge file to describe the connector.

The configuration interface files, placed under `define/applications/` are:

Filename	Description
logiPlexAttributes.xhtml	Base configuration
logiPlexAttributesInclude.xhtml	Additional attribute configuration
logiPlexRulesForm.xhtml	Connector Rules configuration

Note that the default version of logiPlexAttributes.xhtml will work with IdentityIQ 8.0 and later, but not with earlier versions. For IdentityIQ 7.x another version of the file is supplied, called logiPlexAttributes7x.xhtml; you will need to remove logiPlexAttributes.xhtml and rename logiPlexAttributes7x.xhtml to replace it. If you are deploying with SSD v6.0 or 6.1, this will automatically be done for you.

Some sample rules are also provided in the SSD under the folder config/SSF_Tools/LogiPlex_Connector/Samples. These will not be deployed by default.

5.2. PS Extensions (Binary)

The PS Extensions package is a ZIP file that can be overlaid on top of IdentityIQ, or when using the Services Standard Build (SSB) or Services Standard Deployment (SSD) on top of the web/ folder. The XML configuration objects will need to be moved to the config/ folder of the SSB/SSD.

The files included in the distribution are:

File	Description
define/applications/logiPlexAttributesInclude.xhtml	Application settings page (include)
define/applications/logiPlexRulesForm.xhtml	Application rules page
define/applications/logiPlexAttributes7x.xhtml	Application settings page for IdentityIQ 7.x
define/applications/logiPlexAttributes.xhtml	Application settings page for IdentityIQ 8.0 and above
WEB-INF/config/custom/pslabs/connector/logiplex-connector/Configuration/ConnectorRegistry-LogiPlexConnector.xml	Connector Registry entry for the LogiPlex connector in Classic mode. For use in the SSB/SSD, move to config/Configuration to automatically deploy. For stand-alone use, import this file into IdentityIQ through the System Configuration or via the Console.
WEB-INF/lib/logiplex-connector-20200730.0.1.jar	The connector library. The name of the file may differ, depending on the version used.

6. Configuration

6.1. Settings

The connector has just two configuration settings:

Setting	Type	Description
Master Application	Application	The application definition which defines the connector of the real target application
Split Application Prefix	String	If set, every application selected by the split rule on aggregation will be prefixed with this string. If the string does not end in a dash, a dash will be added as a separator. E.g. if the prefix is set to “LGX”, the sub-application “Expenses” will become “LGX-Expenses”. If the prefix is set to “ABC-”, it will become “ABC-Expenses”

Connection settings must be configured on the master application.

6.2. Rules

The connector has two additional rules options. Since we cannot extend the types of rules supported by IdentityIQ, as rule types are defined as an enum, we have to re-use existing rule types.

6.2.1. LogiPlex Split Rule

6.2.1.1. Description

The LogiPlex Split Rule is in fact a rule of type ResourceObjectCustomization, but is run before the actual customization rule is applied. It will receive slightly different inputs and the expected output is also different.

6.2.1.2. Definition and Storage Location

The rule is associated to a LogiPlex application in the UI in the application definition:

Application →Application Definition → select existing or create new application → Rules → LogiPlex Split Rule

6.2.1.3. Arguments

Inputs

Argument	Type	Purpose
object	sailpoint.object.ResourceObject	A reference to the resource object built by the connector
application	sailpoint.object.Application	
applicationName	java.lang.String	The name of the application.
connector	*Not provided*	-
state	java.util.HashMap	A Map that can be used to store and share data between executions of this rule during a single aggregation run.
map	java.util.HashMap <String,List<ResourceObject>>	A pre-prepared map to be filled and returned

Argument	Type	Purpose
util	sailpoint.services.standard.connector.LogiPlexConnector.LogiPlexUtil	An instance of the LogiPlexUtil as used by the iterator, to allow calling convenience methods (see JavaDoc for the connector).

Note: contrary to a normal customization rule, the connector variable is not set.

Outputs

Argument	Type	Purpose
objects	java.util.HashMap	A Map containing application names as keys and corresponding ResourceObject values.

6.2.1.4. Example

This example rule will inspect the account and group objects received from an LDAP server. It will check the name of group memberships. Groups starting with `cn=webapplication` will be related to the **WebApplication** application. Groups starting with `cn=expenses-` will be related to the **Expenses** application. All other groups will be related to the main application. In this example, the choice has been made to also return results for the main application, although this is not required.

```
import sailpoint.tools.Util;
import sailpoint.object.ResourceObject;

public String resolveApplication(String name) {
  if (Util.isNotNullOrEmpty(name)) {
    String lname = name.toLowerCase();
    if (lname.startsWith("cn=webapplication")) {
      return "WebApplication";
    }
    if (lname.startsWith("cn=expenses-")) {
      return "Expenses";
    }
  }
  return application.getName();
}

String applicationName = application.getName();
Map map = new HashMap();

if ("account".equals(object.getObjectType())) {
  List groups = object.getStringList("groups");
  if (groups != null && !groups.isEmpty()) {
    Map groupMap = new HashMap();
    groupMap = util.updateListMap(groupMap, applicationName, null);
    for (String group: groups) {
      String appName = resolveApplication(group);
      if (Util.isNotNullOrEmpty(appName)) {
        groupMap = util.updateListMap(groupMap, appName, group);
      }
    }
    Set keys = groupMap.keySet();
    if (!keys.isEmpty()) {
      for (String key: keys) {
```

```
            List appGroups = groupMap.get(key);
            ResourceObject cloneObject = object.deepCopy(context);
            if (!Util.isEmpty(appGroups)) {
              cloneObject.put("groups", appGroups);
            } else {
              cloneObject.remove("groups");
            }
            map.put(key, cloneObject);
          }
        } else {
          map.put(applicationName, object);
        }
      } else {
        map.put(applicationName, object);
      }
    } else if ("group".equals(object.getObjectType())) {
      String nativeIdentity = object.getIdentity();
      String appName = resolveApplication(nativeIdentity);
      map.put(appName, object);
    } else {
      map.put(applicationName, object);
    }

    return map;
```

In case a large number of groups need to be considered, good alternatives to check for the destination of groups could be:

- A Custom object containing the names or regular expressions to match,
- Additional information on groups in the source application: aggregate groups first to populate the Entitlement Catalog, then reference the entitlement catalog to determine to which application(s) a group should be linked.

6.2.2. LogiPlex Provisioning Rule

6.2.2.1. Description

The LogiPlex Provisioning Rule is in fact a rule of type CompositeRemediation (Logical Provisioning Rule). It accepts a provisioning plan and can modify the plan for the real target application and then return the modified provisioning plan.

If the rule is not provided, the connector will apply internal default logic. This default logic will:

- Clone the plan and included AccountRequest and ObjectRequest objects.
- For each request object targeted at the main application, just keep the request as is.
- For each account request object targeted at a sub application:
    - Handle create operation as create operations only if there is no corresponding account on the main application,
    - Handle create operations as modify operations, applicable only to entitlements, if there already is a corresponding account on the main application,
    - Handle deletion on a sub-application as a removal for all entitlements,
    - Handle changes on sub-application accounts only for entitlements,
    - Handle enabling, disabling, locking and unlocking only on the main application (ignore for sub-applications).

NOTE: When deleting, enabling, disabling, locking or unlocking the main application account, the same operation is applied to sub-application. This is handled *after* the provisioning plan has been executed and applied internally only.

NOTE: Most often, the rule will clone the original plan and modify the included AccountRequest objects. Note that for these objects and the enclosed AttributeRequest objects, it is important to also copy any arguments, as these may be necessary to link information back to its origins after provisioning.

6.2.2.2. Definition and Storage Location

The rule is associated to a LogiPlex application in the UI in the application definition:

Application →Application Definition → select existing or create new application → Rules → LogiPlex Provisioning Rule

6.2.2.3. Arguments

Inputs

Argument	Type	Purpose
identity	`sailpoint.object.Identity`	Reference to the Identity object for whom the provisioning request has been made
plan	`sailpoint.object.ProvisioningPlan`	Reference to a provisioning plan against the LogiPlex application
application	`sailpoint.object.Application`	The LogiPlex application definition
masterApplication	`sailpoint.object.Application`	The master application definition
connector	`sailpoint.connector.AbstractConnector`	An instance of the LogiPlex connector
masterConnector	`sailpoint.connector.AbstractConnector`	An instance of the connector for the master application

Outputs

Argument	Type	Purpose
plan	`sailpoint.object.ProvisioningPlan`	The modified plan

6.2.2.4. Example

The goal of this rule is to merge changes for sub-applications into the main application. It is important to understand that the plan will, after successful provisioning, also be applied to the identity cube. So, the plan must be modified in such a way that whatever will change ends up in the original plan object.

The easiest implementation is by calling the default logic on the main application, which is demonstrated below. After doing that, it is possible to make some additional changes to the plan.

```
import sailpoint.object.ProvisioningPlan;
import sailpoint.services.standard.connector.LogiPlexConnector;

// Expect: sailpoint.connector.AbstractConnector connector

// Use the default logic from the connector to re-compose the plan.
ProvisioningPlan newPlan = ((LogiPlexConnector)
connector).runDefaultProvisioningMergeLogic(plan, plan, identity, true,
masterConnector);

// After using the default logic, optionally, other changes could be applied.

if (log.isTraceEnabled() && newPlan != null) {
  // Dump the plan.
  log.trace(newPlan.toXml());
}
return newPlan;
```

7. Schema Attributes

The schema(s) can either be generated from the master application or entered manually. To generate the schema(s) from the master application, use the **Discover Schema Attributes** for each object type.

8. Provisioning Policies

Provisioning policies will, unfortunately, not be copied from the master application to the LogiPlex application. The provisioning policy or policies will have to be re-configured or copied as XML in the Debug interface.

If the provisioning policy for the master application is setup as a separate Form object, both the master and the LogiPlex application can reference the same Form object.

The field definition for the “identity attribute” and likely also the “display attribute” on the provisioning policy for sub-applications, may need to be adjusted such that it will not generate a new value, but look for an existing account on the main application. If an account on the main application is found, the values for the “identity attribute” and “display attribute” should be copied from the main application account.

Installation and Configuration – Adapter Mode

9. Implementation Overview

The implementation of the LogiPlex connector in adapter mode require the following steps:

- Deployment of the connector,
- Set up the master application and confirm connectivity,
  - Test the configuration and schema,
  - Define and test provisioning.
- Modify the master application XML to become a LogiPlex application:
  - Replace the connector class with the LogiPlex connector class,
  - Set the original connector class in the attribute `logiPlexMasterConnector`.
  - Optionally define a prefix for auto-generated sub-applications,
  - Define a split rule to determine which entitlements belong to which sub-application
  - Optionally define a merge rule
  - For generated sub-applications:
    - Manually copy configuration options,
    - Manually copy and update the provisioning policy.

10. Installation Files

10.1. Services Standard Deployment (6.x and below)

If you are deploying IdentityIQ 7.2 or greater with the Services Standard Deployment (SSD), the LogiPlex Connector will be deployed by default, and there are no extra steps to take to install it. Deployment is controlled by this property in the `build.properties` file:

```
deployLogiPlexConnector=true
```

Setting it to false will prevent the connector being deployed.

The IdentityIQ LogiPlex Connector consists of the following Java class files:

Filename	Description
`LogiPlexConnector.java`	Main LogiPlex Connector Java class

Other files (see chapter 5) will also be deployed as part of an SSD deployment but are not needed when adapter mode is used.

Some sample rules are also provided in the SSD under the folder `config/SSF_Tools/LogiPlex_Connector/Samples`. These will not be deployed by default.

10.1. PS Extensions (Binary)

The PS Extensions package is a ZIP file that can be overlaid on top of IdentityIQ, or when using the Services Standard Build (SSB) or Services Standard Deployment (SSD) on top of the `web/` folder. The XML configuration objects will need to be moved to the `config/` folder of the SSB/SSD.

The files included in the distribution are:

File	Description
`define/applications/logiPlexAttributesInclude.xhtml`	Application settings page (include)
`define/applications/logiPlexRulesForm.xhtml`	Application rules page
`define/applications/logiPlexAttributes7x.xhtml`	Application settings page for IdentityIQ 7.x
`define/applications/logiPlexAttributes.xhtml`	Application settings page for IdentityIQ 8.0 and above
`WEB-INF/config/custom/pslabs/connector/logiplex-connector/Configuration/ConnectorRegistry-LogiPlexConnector.xml`	This file can be discarded when using Adapter mode.
`WEB-INF/lib/logiplex-connector-20200730.0.1.jar`	The connector library. The name of the file may differ, depending on the version used.

11. Configuration

11.1. Settings

In adapter mode, all settings have to be changed in the XML directly, through the debug interface, an exported XML document, or preferably in the IdentityIQ Deployment Accelerator.

11.1.1. Connector Class

The most important change is the connector class. The following changes are needed:

- In the <Application> tag, change the connector class to the LogiPlex connector class.
- Put the original connector class in the `logiPlexMasterConnector` attribute.

Original (example: Active Directory):

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE Application PUBLIC "sailpoint.dtd" "sailpoint.dtd">
<Application authoritative="false" connector="sailpoint.connector.ADLDAPConnector"
featuresString="PROVISIONING, SYNC_PROVISIONING, AUTHENTICATE, MANAGER_LOOKUP,
SEARCH, UNLOCK, ENABLE, PASSWORD, CURRENT_PASSWORD" icon="directory1Icon"
name="Active Directory" profileClass="" type="Active Directory - Direct">
  <Attributes>
…
```

Updated:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE Application PUBLIC "sailpoint.dtd" "sailpoint.dtd">
```

```
<Application authoritative="false"
connector="sailpoint.services.standard.connector.LogiPlexConnector"
featuresString="PROVISIONING, SYNC_PROVISIONING, AUTHENTICATE, MANAGER_LOOKUP,
SEARCH, UNLOCK, ENABLE, PASSWORD, CURRENT_PASSWORD" icon="directory1Icon"
name="Active Directory" profileClass="" type="Active Directory - Direct">
  <Attributes>
…
      <entry key="logiPlexMasterConnector"
value="sailpoint.connector.ADLDAPConnector"/>
…
```

11.1.2. LogiPlex Connector Settings

The following settings should be edited in the XML definition. All of these go into the Attribute section of the application definition.

Attribute	Type	Required	Description
logiPlexMasterConnector	String	Yes	The full class name of the original connector to be used by the LogiPlex connector.
logiPlexPrefix	String	No	If set, every application selected by the split rule on aggregation will be prefixed with this string. If the string does not end in a dash, a dash will be added as a separator. E.g. if the prefix is set to “LGX”, the sub-application “Expenses” will become “LGX-Expenses”. If the prefix is set to “ABC-”, it will become “ABC-Expenses”
logiPlexAggregationRule	String	Yes	The name of the rule that will be used during aggregation (the “split rule”) to determine to which sub-application entitlements belong.
logiPlexProvisioningRule	String	No	Optional: the name of the rule that will be used to modify the provisioning plan (the “merge rule”). If no rule is configured, the default behavior will be used.

Attribute	Type	Required	Description
`logiPlexUseGetObject`	boolean	No	If set to true, on creation of an account, the connector will perform a single object aggregation to check whether a (main) account exists when creating a new account, if it cannot find a Link object on the cube. This feature allows simultaneous requests for multiple sub-applications even if the user does not have a main/master account, yet. This only works if the connector supports single object aggregations. The behavior can be overridden using a provisioning rule.
`logiPlexSimulateProvisioning`	boolean	No	If set to true, provisioning will not take place, but only the processing of the plan will be performed. This is useful when debugging the provisioning logic.
`logiPlexSimulateProvisioningResult`	String	No	If logiPlexSimulateProvisioning is set to true, this option controls the result code that is returned from simulation. Valid values are: • committed • failed • queued • retry

11.2. Rules

The rules in the adapter mode are the same as used in classic mode. See §6.2.

12. Schema Attributes

The schema(s) should be setup before changing the application into a LogiPlex application. After that, the application schemas can still be modified as usual. If the master application supports discovering the schema, the **Discover Schema Attributes** button can be used for each object type. The request will simply be forwarded to the master connector.

13. Provisioning Policies

Provisioning policies will, unfortunately, not be copied from the master application to the LogiPlex application. The provisioning policy or policies will have to be re-configured or copied as XML in the Debug interface.

If the provisioning policy for the master application is setup as a separate Form object, both the master and the LogiPlex application can reference the same Form object.

14. Cloud Gateway Support

Cloud Gateway support requires version 2020.07.07-0001 or above of the connector code. Since this version, aggregation and provisioning via the Cloud Gateway is possible. Cloud Gateway support is tested and supported only in Adapter Mode.

There are two strategies that can be used: one strategy (supported since version 2020.07.09-0003) requires an attribute to be configured on the master/main application. The LogiPlex connector will then call the Cloud Gateway connector, instead of the real connector. The LogiPlex connector operates only on IdentityIQ, while the Cloud Gateway only executes the real connector.

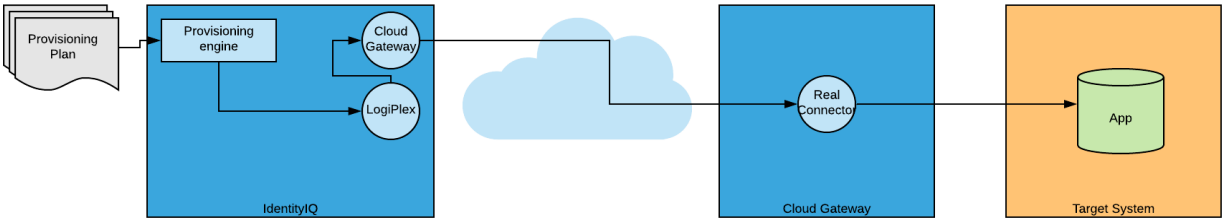


Figure 3: Strategy 1, Cloud Gateway initiated by LogiPlex

The other strategy (supported since version 2020.07.07-0001) needs the Cloud Gateway application to be configured as the proxy on the master/main application. This can be done in the UI but is a less stable option and more complex as it requires the identity to be sent along with request and the LogiPlex connector must be installed on both IdentityIQ and the Cloud Gateway.

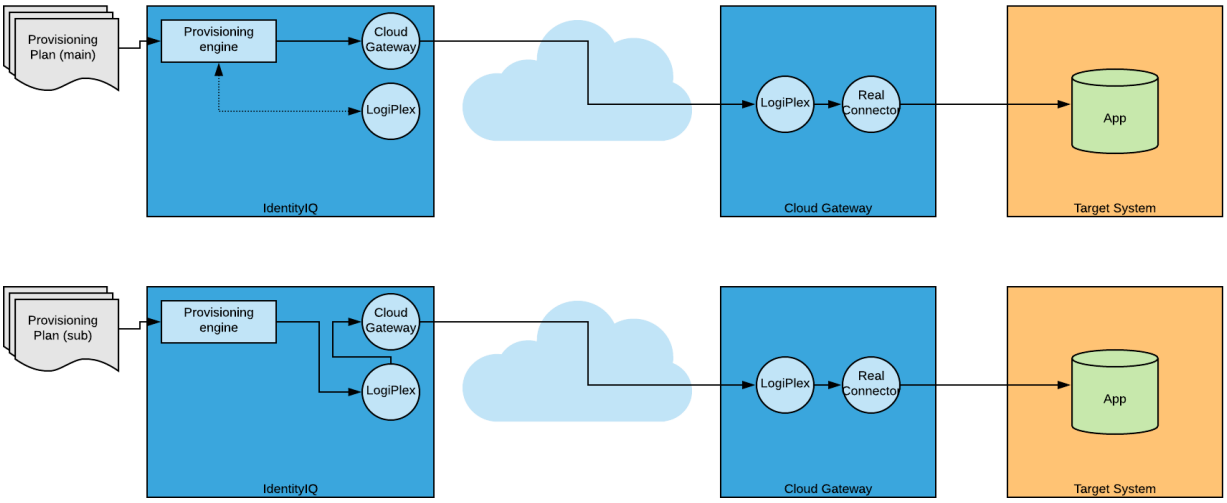


Figure 4: Strategy 2, Cloud Gateway partly initiated by provisioning engine

14.1. Cloud Gateway as XML Attribute

On the master/main application, there must not be a proxy defined.

In the XML, however, an extra attribute is required that points to the Cloud Gateway application:

```
<entry key="logiPlexAggregationRule" value="389DS LogiPlex Split Rule"/>
<entry key="logiPlexCloudGatewayApplication" value="Cloud Gateway"/>
<entry key="logiPlexMasterConnector" value="sailpoint.connector.LDAPConnector"/>
<entry key="logiPlexPrefix" value="LPX"/>
<entry key="logiPlexSimulateProvisioning" value="false"/>
<entry key="logiPlexSimulateProvisioningResult" value="failed"/>
<entry key="logiPlexUseGetObject" value="true"/>
```

The attribute `logiPlexCloudGatewayApplication` must specify the name of the Cloud Gateway application. This value is used to set up the master connector when the LogiPlex connector is initiated and to perform a couple of additional checks when the connector to the target system is set up for aggregation or provisioning.

14.1.1. Synchronization Task

Once the master/main application has been prepared, a dedicated synchronization task must be run, to ensure that the configuration is synchronized to the Cloud Gateway. This task must be run in addition to the out-of-the-box Cloud Gateway synchronization task.

The task is called “LogiPlex Cloud Gateway Synchronization Task” and has no configuration options. It will look for any applications that use the LogiPlex connector, do not have a proxy configured and have

the attribute `logiPlexCloudGatewayApplication` specified. It will set up the Cloud Gateway connector and process the application definition:

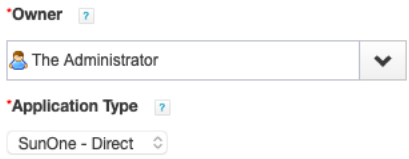
- Create an in-memory copy of the application definition,
- Replace the LogiPlex connector with the real connector,
- Strip any LogiPlex-specific attributes.

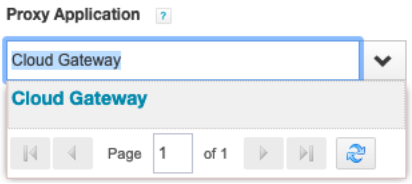
Then it sends the application definition to the Cloud Gateway.

Important: make sure to **exclude** the LogiPlex applications from the out-of-the-box Cloud Gateway synchronization task, as this task will not make the required changes. Schedule both synchronization tasks to be run one after another.

14.2. Cloud Gateway as Proxy

On the master/main application, the Cloud Gateway application must be configured as the proxy.





The sub-applications must have the master/main application as their proxy, which is automatically done when the sub-applications are generated. Under the hood, in the XML, the master/main application must also be configured in a special attribute:

```
<entry key="logiPlexCGWProxyApplication" value="389DS"/>
```

14.2.1. Limitations

With this approach there are some limitations and things to keep in mind:

- Cloud Gateway support is tested only in Adapter mode.
- Partitioned aggregation is not supported.
- Aggregation **must** always be performed through the main application.
- The LogiPlex connector must be installed on both the IdentityIQ servers and the Cloud Gateway. Both must be the exact same version.
- The Cloud Gateway must be configured on the Main/Master application only. If the Cloud Gateway is configured as the proxy for sub-applications this won't work as expected and LogiPlex aggregation will reset the proxy to the main application anyway.
- The synchronization task must be configured to synchronize all the sub-application definitions to the Cloud Gateway. The use of the LogiPlex connector can, due to its multiplex-based nature, automatically generate application definitions. In case of the Cloud Gateway, a proxy generator rule[1] should be used, which can – apart from filling in the necessary details for the application definition – update the synchronization task with the newly created application.

[1] https://community.sailpoint.com/t5/IdentityIQ-Wiki/LogiPlex-Sub-Application-Creation/ta-p/76964

- The synchronization task will strip the proxy application from any application definition that is sent to the Cloud Gateway. This also means that the link between the main and sub-applications is lost. To allow the LogiPlex connector to find the main application again, an extra attribute must be added to the application's attributes: `logiPlexCGWProxyApplication`. The value must be the name of the main application.

```
<entry key="logiPlexCGWProxyApplication" value="389DS"/>
```

- Provisioned changes may require a full re-aggregation in order to be reflected correctly on the identity cube. This is due to some after-provisioning logic that is not completely taken into account when running on the Cloud Gateway: the plan modified by the connector on the Cloud Gateway is not used to update the identity cube.

14.2.2. Identity "Smuggling"

The LogiPlex connector's default provisioning logic requires knowledge about the current state of the identity for which changes are being handled. On the Cloud Gateway, however, the identity cannot be looked up and the plan will only contain the name of the identity object.

To solve this, the identity object needs to be "smuggled" to the other side. The mechanism to do this would be the provisioning plan. The identity object itself cannot be embedded as an attribute of the plan. The identity is a complex object and embedding this will result in JSON encoding errors. The object needs to be represented as a simple, easy to transport String object.

The trick to do this, is to first get the XML representation of the identity object and use Base64 encoding to get a string that can be safely embedded into the plan. The connector on the gateway can then reverse this process: decode the Base64 string into an XML representation of the object. Then re-create the object back from the XML. This "smuggling" can be prepared in the Before Provisioning Rule, which must be configured on the main and all sub-applications.

The following is an example rule to embed the identity:

```
import sailpoint.object.Identity;
import sailpoint.tools.Util;

String nativeIdentity = plan.getNativeIdentity();
if (Util.isNotNullOrEmpty(nativeIdentity)) {
  Identity identity = context.getObjectByName(Identity.class, nativeIdentity);
  if (identity != null) {
    plan.setIdentity(identity);
    String encodedXml =
Base64.getEncoder().encodeToString(identity.toXml().getBytes());
    plan.put("logiPlexEmbeddedIdentity", encodedXml);
  }
  log.error(plan.toXml());
}
```

The before provisioning rule must be forced to run on IdentityIQ. If this is not configured, it will run on the gateway and thus fail. To force the rule to run on IdentityIQ, add the following to the application attributes of the main application and all sub-applications:

```
<entry key="beforeProvisionRuleLocation" value="proxy"/>
```

14.2.3. Connector Rule Execution

LogiPlex-specific rule (connector rules) will be executed on the Cloud Gateway. This concerns the split rule and optionally the provisioning merge rule. Rules executed on the Cloud Gateway do not have access to the IdentityIQ database but rely entirely on information that has been synchronized to the Cloud Gateway. Currently, the only object types that can be synchronized by the synchronization task to the Cloud Gateway are applications, rules and the system configuration (hidden setting).

This means that any rules executed on the Cloud Gateway cannot look up an identity object, or check accounts (Links), roles, entitlements, etc.

For provisioning, whenever possible, it is highly recommended to use the default provisioning logic, as this will make use of the “smuggled” identity.

14.2.4. Cloud Gateway Support Inner Workings

When IdentityIQ figures out how to provision, it will:

- Check whether there is any integration module or application that acts like an integration module that explicitly “manages” the application of the account request. If so, all provisioning requests will be redirected to that integration module or application.
- Check whether the application supports provisioning and if so, check whether the application specifies another application as a proxy.
- If the application does not support provisioning by itself and neither does its proxy, the next check is for any integration module that is marked as the “universal manager”.
- If there is no “universal manager”, a work item is created to keep track of manual provisioning.

If a proxy is found, the proxy will be instantiated to perform provisioning. In case of a LogiPlex sub-application, there will in this case, be two levels of proxies:

Sub-application → Main application → Cloud Gateway.

Resolving, however, takes only one level into account. So, when provisioning access for a sub-application, IdentityIQ will instantiate the LogiPlex connector for the main application, but not the Cloud Gateway. The LogiPlex connector will figure out that the main application has the Cloud Gateway configured and instead of instantiating the “real” connector, it will instantiate the Cloud Gateway connector and send the provisioning plan through that connector to the gateway.

The gateway launches the LogiPlex connector, which will in turn use the real connector on the Cloud Gateway to do provisioning.

15. Utility Methods

15.1. runDefaultProvisioningMergeLogic

This method is a method on the connector class that can be used in the LogiPlex provisioning rule. It takes the original plan and a pre-prepared new plan, plus an identity and will apply pre-defined logic to the plan to handle changes to the main and sub-applications. If not provisioning rule is defined, this logic will be applied out of the box by the connector itself.

This method has two instances:

```
public ProvisioningPlan runDefaultProvisioningMergeLogic( ProvisioningPlan plan,
ProvisioningPlan originalPlan, Identity identity) throws GeneralException {


public ProvisioningPlan runDefaultProvisioningMergeLogic( ProvisioningPlan plan,
ProvisioningPlan originalPlan, Identity identity, boolean useGetObject, Connector
masterConnector) throws GeneralException {
```

The first instance simply calls the second instance with the provided parameters, plus `useGetObject = false` and `masterConnector = null`.

Parameter	Type	Description
plan	sailpoint.object.ProvisioningPlan	The provisioning plan to update.
originalPlan	sailpoint.object.ProvisioningPlan	The original provisioning plan (can be the same a plan in most cases).
identity	sailpoint.object.Identity	The identity for whom provisioning is being performed.
useGetObject	boolean	For provisioning, this option enables contacting the target system – if supported – to verify account existence in order to make the correct decision about account creation.
masterConnector	sailpoint.connector.Connector	An instance of the master connector to use for reading directly from the target system.

16. Known Issues

16.1. Test Connection

When clicking the Test Connection button in “classic mode”, an error will appear along with a success message.

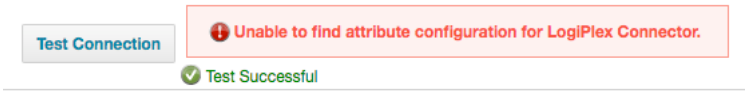


Figure 5: Error during Test Connection

This specific error message can be ignored. The connector works just fine. Other errors may indicate an issue with the setup or the configuration of the *Master* application.

16.2. Aggregation of Sub-Applications

Like with a multiplex application, aggregation should be performed only on the main application. Aggregating directly from a sub-application may not have the desired outcome, depending on how rules are setup.

16.3. Partitioning

Dividing partitions only works correctly if the connector natively supports partitioning. Application level partitioning will work, but the numbers used to divide the partitions will be off. When calculating the number of partitions, the master application will return the raw number of accounts. When the split rule is applied, the number of accounts may be a lot higher.

As shown in the picture below, the number of aggregated accounts is already higher than the expected number of accounts per partition. In this example, we are aggregating a delimited file with 100,000 lines, corresponding to 100,000 accounts.

If partitioning is configured to create 4 partitions, like in this example, it expects to aggregate 25,000 accounts per partition. The LogiPlex Split Rule may return multiple accounts (main and sub) for each line read.

Partitioned Results

Name	Host	Status
Fake Names LogiPlex - Accounts 1 to 25000	Akira	Refreshing 28860 Hismay
Fake Names LogiPlex - Accounts 25001 to 50000	Akira	Refreshing 27610 Higend
Fake Names LogiPlex - Accounts 50001 to 75000	Akira	Refreshing 27614 Bobjew
Fake Names LogiPlex - Accounts 75001 to 100000	Akira	Refreshing 27655 Goodgicess60
Partitioning	Akira	Success
Finish Aggregation		Waiting

Page 1 of 1 Displaying 1 - 6 of 6

Figure 6: Results for Application Level Partitioning

16.4. Delta Aggregation

Depending on the strategy for delta aggregation used by the connector, it may need to store information on the application definition. For example, for a directory server (LDAP) where a changelog container is used (`cn=changelog`), the connector will store the latest processed change number for accounts and groups to the application definition, so it knows where to pick up changes on the next delta aggregation.

For delta aggregation, the application definition for the master connector must be updated to keep that information, which will not succeed in “classic mode” due to the fact that the application that the “Aggregator” uses is the LogiPlex main application.

Customers who require delta aggregation with LDAP, Active Directory and other application types that need delta information to be stored in the application definition, should switch to “adapter mode”. In that case, there will only be one application definition involved, which is correctly updated after each aggregation with delta information.

16.5. Version 7.1 versus 7.2+

There is minor difference between version IdentityIQ 7.1 and 7.2 that requires a small change in the connector code when moving between the versions. The class `ExpiredPasswordException` has been moved from the package `sailpoint.api` to `sailpoint.connector` in 7.2 and up.

So, IdentityIQ 7.1 needs this import:

```
import sailpoint.api.ExpiredPasswordException;
```

And IdentityIQ 7.2 and higher versions needs this import:

```
import sailpoint.connector.ExpiredPasswordException;
```

If you are using the SSD to create your build, the source code modification will be done by the build scripts and you will not need to change anything. The PS Labs provided versions (2020.07.30-0001 and up) are binary builds and only support IdentityIQ 7.2 and above.

17. Do's and Don'ts

17.1. Mode

The connector can operate in two modes: Classic Mode (3.4.1) and Adapter Mode (3.4.2).

When starting a new project, the recommended mode is the Adapter Mode. This mode has a number of benefits over the Classic Mode:

- There is no need to set up a separate main and master application.
- There is no more duplication of entitlements and accounts.
- Adapter mode has better support for delta aggregation, especially for Active Directory and other application types, where delta information must be stored with the application definition.

The only downside is that it cannot be configured through the UI, but modifications need to be applied through XML.

17.2. Nested Groups

Directories often support groups being a member of other groups, in other words, nested groups. IdentityIQ only understands groups being nested within one single application. If groups are nested, the nesting will only be recognized if all groups (top-level and member groups) belong to the same application, main or sub. IdentityIQ will not recognize the nested groups if the top-level and member groups are spread across main and/or sub-applications.

17.3. Secondary Accounts

In (for example) directory servers like Active Directory, it is not uncommon that people have multiple accounts, like a "normal user" account, a developer account and an administrative account.

When access is requested for a secondary account on a sub-application that does not yet exist on the main application, this may lead to confusion. A few options to help with this situation:

- Use of roles with account selectors to encapsulate such access.
- A custom workflow to request the secondary account first, before requesting other.
- A custom workflow specifically to manage access on secondary accounts.
- A mechanism to "recognize" a secondary account through a dummy entitlement, so that the secondary account can be selected when requesting access.

Possibly a combination of the above is needed.

17.4. Exchange, Lync/Skype for Business, etc.

The main access to Exchange, Lync/Skype for Business and other features often managed through Active Directory should be primarily tied to the main account. Certain group memberships may be related to sub-applications.